

define her set of bad times. Periods of low wealth are often bad times, but bad times also include times of moral indignation, jealousy when her peers and neighbors do better, and distress at how her wealth and consumption might have changed for the worse compared to what she was used to. Asset management is ultimately about building portfolios that mitigate the risk of an investor's bad times, while trading off the pain of those bad times for the potential of returns.

We have whiz-bang methods of handling bad times that arise from high volatility and low average returns (mean-variance optimization), but generally our quantitative methods of handling more general varieties of bad times leave much to be desired. This is especially true of modeling how bad times change over investors' life cycles, and dynamically handling bad times of illiquidity over long horizons, incorporating time-varying liabilities, and taking into account asymmetric treatments of losses and gains. Progress is being made, but it would be nice if we had a standard commercial optimizer that enabled us to toggle between various utility functions, especially those incorporating downside risk aversion, and see the effects on optimal portfolios. Investors need help in mapping a series of bad times and how they perceive the risk of bad times to different quantitative models of investing.

Factor Risk Premiums Are Rewards for Suffering Losses during Bad Times

The concept of factor investing compares the bad times of an investor with bad times of the typical investor—the market investor. In fact, since the market portfolio is strictly a long-only passive holding of assets, the market investor does not capture any dynamic factor risk premiums. The market also does not rebalance. We can move away from market weights to holding portfolios of securities which outperform, on average. These high returns don't come for free. Factor portfolios earn risk premiums to compensate investors for bearing losses during bad times. Bad times for the typical investor, however, may not be a bad time for you. This gives you scope to harvest a factor risk premium if you can tolerate the losses. Other investors willing to give up the factor exposure because they are too risk averse.

We are seeing many innovations that customize factor indexes with nonmarket capitalization weights, tailor the size of the risk exposures, and more cheaply create optimal factor portfolios. These are all good things. But when asset owners choose between different factor risks, and the size of the risk exposures, they rarely ask themselves: how am I different from the market? Ideally, consultants, financial advisors, and asset managers would offer guidance on how factor risk losses correspond to different bad times. We should know some economics explaining the existence of these factor risk premiums—even for the basic equity premium—before we can say whether they are appropriate for an investor. Understanding the economic theories giving rise to risk premiums, relating these stories to the realized losses and gains of various factors, and showing how the factor risks are appropriate (or not) for a particular investor is the optimal